

WISER × MODERNA QUANTUM CHALLENGE 2026

RNA Secondary Structure Prediction via Quantum–Classical Hybrid Optimization

QUBO formulation • QAOA / VQE / Interference+Greedy solvers • Hierarchical divide-and-conquer scaling

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The Biological and Computational Problem

Why RNA folding matters

mRNA secondary structure governs molecular stability, translation efficiency, and manufacturability of mRNA therapeutics.

Predicting the minimum free energy (MFE) fold is a core input to rational mRNA design.

The number of possible folds grows exponentially with length — exhaustive classical enumeration is intractable well before practical sequence lengths.

QUBO formulation

- Decision variable: one binary variable per chemically valid candidate base pair (i,j)
- Diagonal terms: real ViennaRNA loop-closure energy per pair
- Off-diagonal terms: stacking-cooperativity bonus between adjacent nested pairs — required, not optional: omitting it collapses the optimum to the unfolded structure
- Constraints: one-pair-per-base and no-pseudoknot penalties, applied strictly as cross-terms (only on actual conflict)

Quantum Approaches Evaluated

1

QAOA

Quantum Approximate Optimization Algorithm,
p = 2 layers

2

VQE

Variational Quantum Eigensolver, hardware-
efficient ansatz

3

Interference+Greedy

Hadamard transform + phase encoding + greedy
refinement

Benchmark: 10-nt Challenge Sequence Prefix

10-nt prefix (GGAGCAAAAC) — ViennaRNA MFE is unfolded, all bases unpaired, $E = 0.000$ kcal/mol

Solver	Gap (kcal/mol)	F1	Runtime (s)
Brute Force	0.000	1.00	0.000
QAOA	0.000	1.00	1.40
VQE	3.000	0.00	0.53
Interference+Greedy	0.000	1.00	1.42

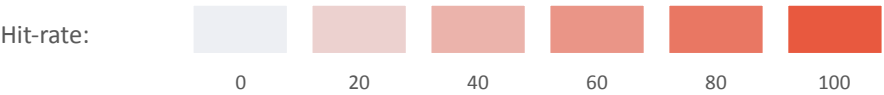
Key finding

QAOA and Interference+Greedy both recover the exact MFE structure (F1 = 1.00, gap = 0.00).

VQE does not — consistent with a barren-plateau-style trainability barrier in the RY+CNOT ansatz family, isolated separately by comparing against random parameter search and an all-to-all entangling topology.

Solver Comparison: Does Quantum Interference Add Value?

	Random Greedy	QAOA	VQE	Interference +Greedy
GCGCAUGC GC	20	100	0	100
GGGAUAUCCC	40	0	0	100
AUGCGCAUGC	100	100	0	100
GGCGCAAGCGCC	0	0	0	100
GCUGCAAAGCU	0	100	0	100
ACGGGCCACCG	20	100	0	100
GCGGCCACGCUA	0	0	0	100
GCCGUGAAGCGG	20	0	0	100



Reading the grid

Each row is one test sequence; each column is one solver. Darker = closer to the brute-force optimum.

Interference+Greedy is solid dark across every row — 8/8. **VQE** is blank across every row — 0/8.

Interference+Greedy: 8/8 exact matches across every sequence tested. QAOA: 4/8 (single stochastic trial). VQE: 0/8.

44-nt Challenge Sequence: Exact Recovery

Sequence: GGAGCAAAACUUGUCGAUUGAGAACAAAUAACAGAAUUUGCUUG

MFE / Solver: .(((((((..((((...(((.....)))...)))..)))))). (both identical)

E = -7.90 kcal/mol, both ViennaRNA MFE and hierarchical solver

0.00

Energy Gap (kcal/mol)

1.00

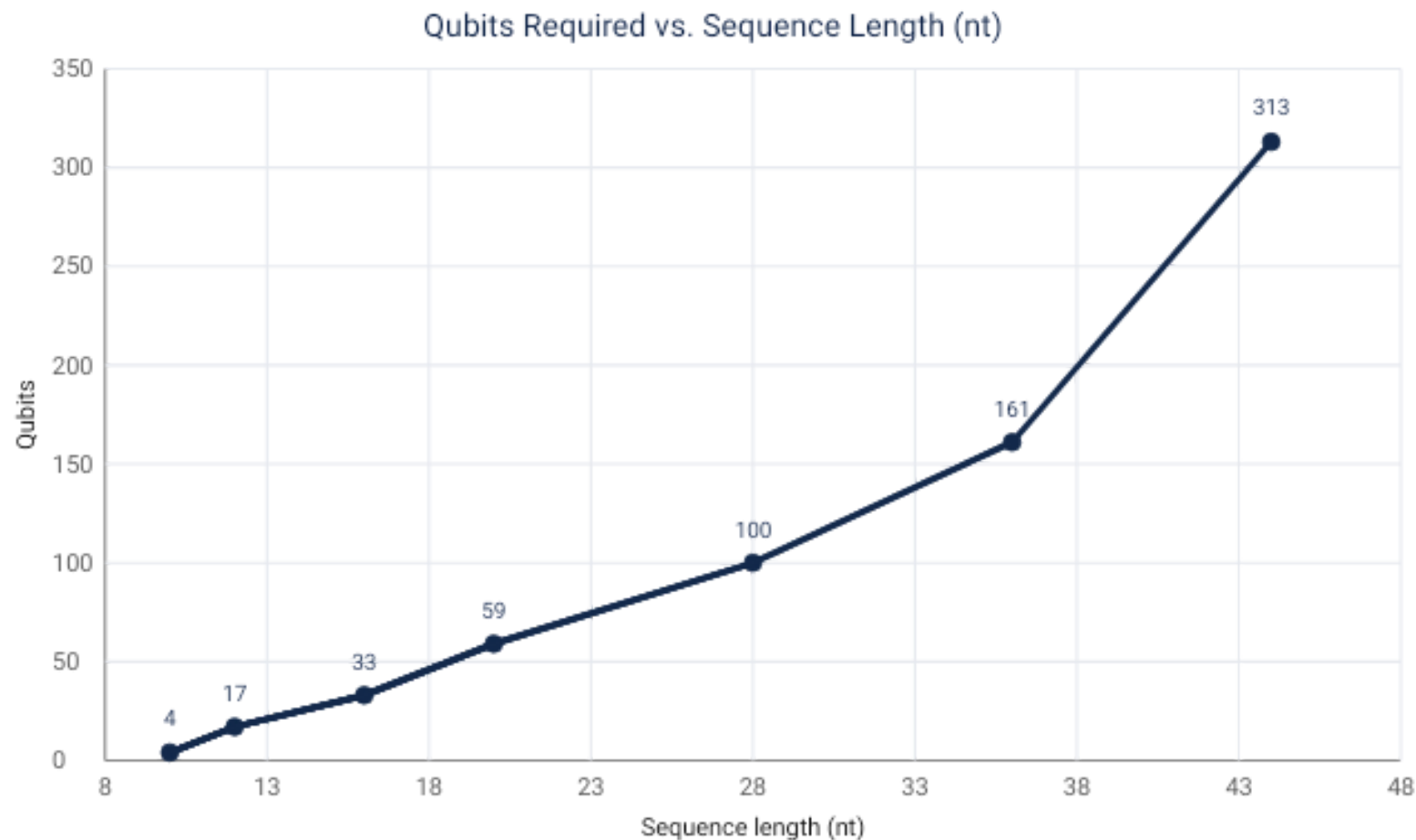
F1-Score (exact match)

313 → 16

Qubits: direct → per subproblem

The recovered structure contains three nested helical regions enclosing a central interior loop — exact recovery of a non-trivial multi-helix topology, solved with at most 16 qubits per subproblem via hierarchical divide-and-conquer.

Quantum Resource Scaling

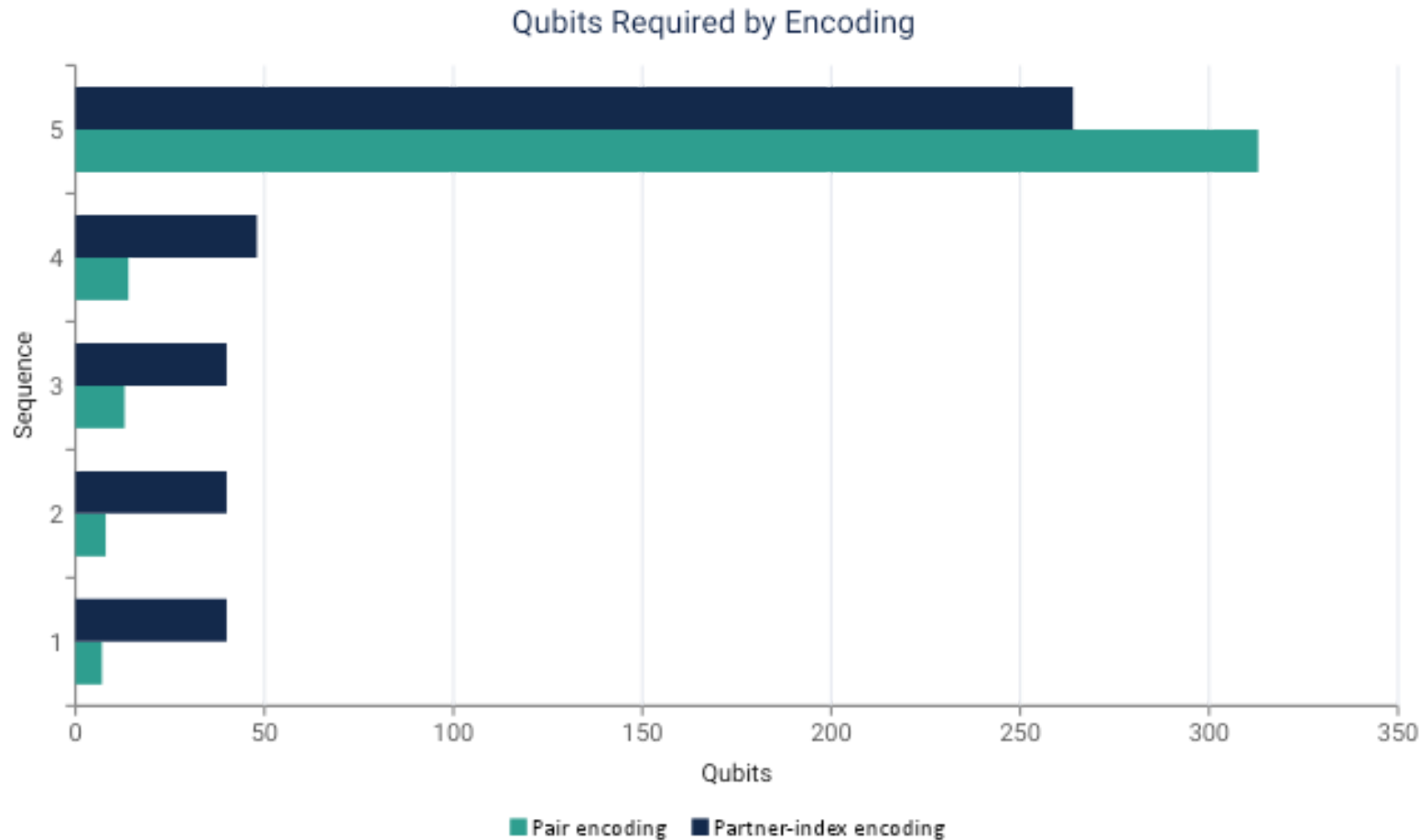


313

qubits required for the
44-nt challenge sequence

Scaling is roughly quadratic under dense
pairing — far exceeding current NISQ
device capacity for exact statevector
simulation.

Encoding Comparison



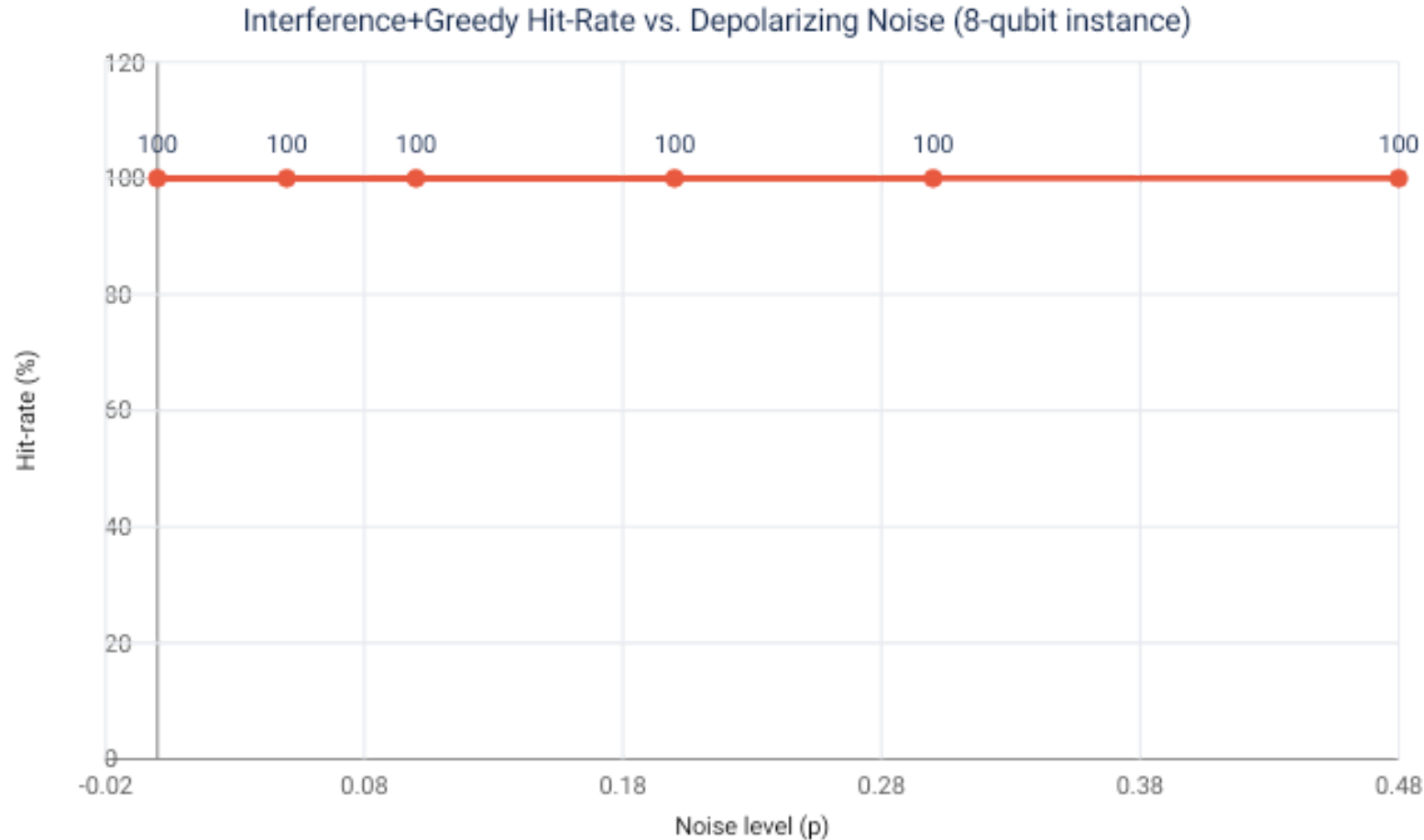
Helix-Level Encoding

Groups contiguous stacked pairs into helical runs.

40–75% fewer qubits than the pair-level encoding (e.g. 14 → 6 qubits on a 14-nt sequence).

Used by the hierarchical solver to reach the 44-nt result.

Noise Robustness



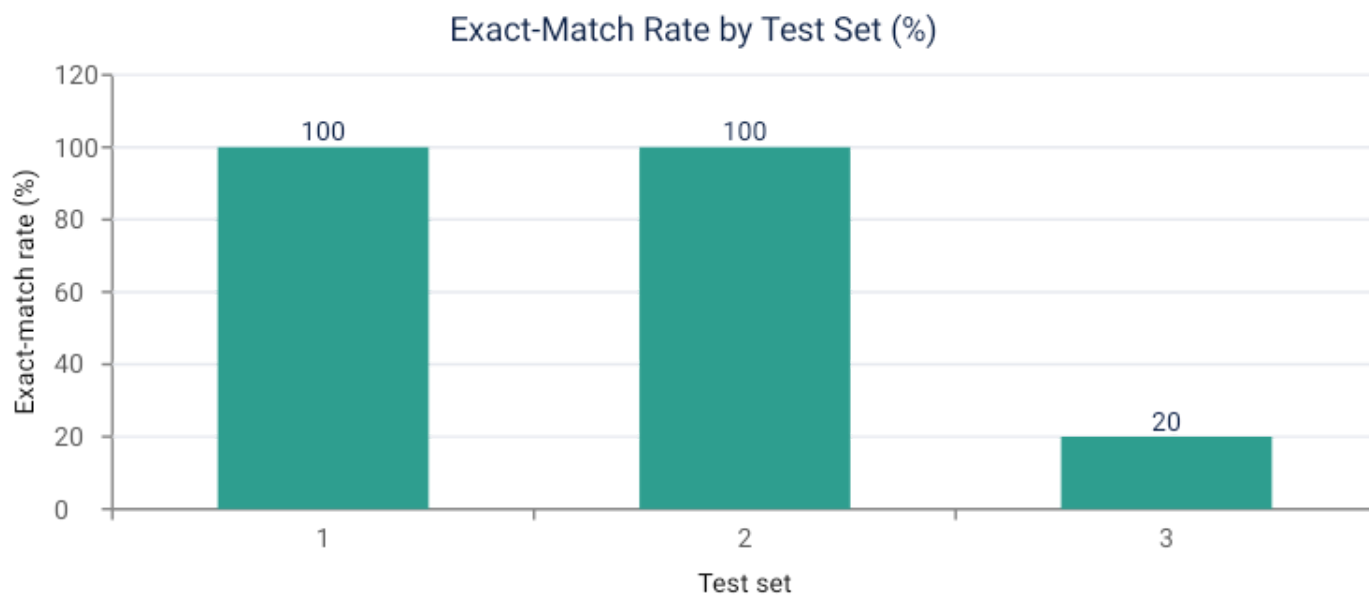
Exact recovery maintained

at every tested noise level, up to $p = 0.50$, on this instance.

Resource constraint:

Density-matrix simulation scales as $O(4^m)$ memory vs. $O(2^m)$ for pure states, restricting exact noise studies to small instances ($m \leq 10$ here).

Hierarchical Divide-and-Conquer Solver

**Limitation:**

Bounded top-K anchor search is a heuristic, not exhaustive backtracking. A suboptimal choice at one level can foreclose the correct continuation at a deeper level. A general fix requires exhaustive backtracking or a DP recursion over all split points (Nussinov/Zuker).

Simulator Backend and Future Directions

Tools and Technologies

Backend	QAOA (60 iters)	Interference+Greedy (3000 shots)
default.qubit	4.49 s	0.039 s
lightning.qubit	1.55 s	0.010 s
Speedup	2.9×	3.9×

- PennyLane (lightning.qubit simulator)
- ViennaRNA (classical RNA folding)
- SciPy (COBYLA)

Future Directions

- Hardware execution: validate the interference decoder on real quantum hardware
- Error mitigation: zero-noise extrapolation, probabilistic error cancellation
- Full Turner energy model: internal loops, bulges, terminal mismatches
- DP integration: exact recursion over split points to guarantee optimality
- Larger sequences: 100–200 nt scalability testing

CONCLUSION

Quantum-classical hybrid optimization reproduces classical MFE RNA structures with high fidelity on every tested instance.

Interference+Greedy outperforms the variational approaches (QAOA, VQE) on this problem class, matching the brute-force optimum on all 8 solver-comparison sequences. The hierarchical divide-and-conquer solver bridges the scaling gap, recovering the exact structure of the 44-nt challenge sequence despite a 313-qubit direct requirement.

8/8

Interference+Greedy exact matches

12/12

Base validation set

1.00

44-nt F1-score

40-75%

Helix-encoding reduction

Interference-based quantum heuristics, combined with classical refinement and problem decomposition, are a practical direction for this problem class on near-term hardware.